

Here are the connections in table form, in the order you'd build them.

1. Sensors and LCD to the ESP32

Component	Component pin	ESP32 pin	Notes
LCD (I2C)	GND	GND	Shared ground rail
	VCC	VIN (5V)	Needs 5V to be readable
	SDA	GPIO 21	I2C data
	SCL	GPIO 22	I2C clock
I2C light sensor	GND	GND	Shared ground rail
	VCC	3V3	
	SDA	GPIO 21	Same pin as the LCD, the bus is shared
	SCL	GPIO 22	Same pin as the LCD
DS18B20 probe	Black (GND)	GND	
	Red (VCC)	3V3	
	Yellow (data)	GPIO 4	Use <code>INPUT_PULLUP</code> in code, or a 4.7k Ω resistor to 3V3 if it reads -127°C
DHT11	GND (-)	GND	
	VCC (+)	3V3	
	DATA (S)	GPIO 18	
Moisture sensor	GND	GND	
	VCC	3V3	Never 5V, or the analog output can exceed 3.3V
	AO (analog out)	GPIO 34	Input-only ADC pin
LDR module	GND	GND	
	VCC	3V3	
	AO (analog out)	GPIO 35	Input-only ADC pin

If your LDR is a bare two-leg part rather than a module, it can't be wired this way. It needs a resistor to form a voltage divider.

2. Battery, charger and boost converter

From	To	Wire
18650 holder red (+)	TP4056 B+	Soldered
18650 holder black (-)	TP4056 B-	Soldered
TP4056 OUT+	MT3608 IN+	Positive
TP4056 OUT-	MT3608 IN-	Ground
MT3608 OUT+	ESP32 VIN	Only after setting 5.0V
MT3608 OUT-	ESP32 GND	Joins the shared ground rail
USB-C charger	TP4056 Type-C port	Charging only

3. Which port to use, and when

Mode	What's plugged in	What must be disconnected
Programming	Laptop USB to the ESP32 USB port	MT3608 output wire from VIN
Charging the cell	Charger to the TP4056 Type-C port	Nothing, the ESP32 can be on or off
Standalone running	Battery chain to ESP32 VIN	USB cable from the ESP32

4. Power rails summary

Rail	Powers	Never connect to
VIN (5V)	LCD only	Any sensor, any GPIO
3V3	DS18B20, DHT11, light sensor, moisture, LDR	The MT3608 output or the battery directly
GND (shared rail)	Everything	

Pins to avoid: don't use GPIO 12 and 13 (they caused your earlier upload failure), and avoid GPIO 0, 2, 5, 6-11, 15.

If you'd like, I can write the single sketch that reads all these sensors and cycles the values across the LCD.